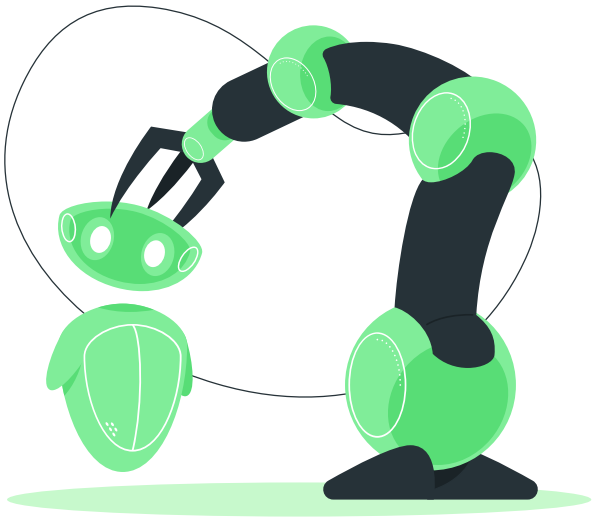
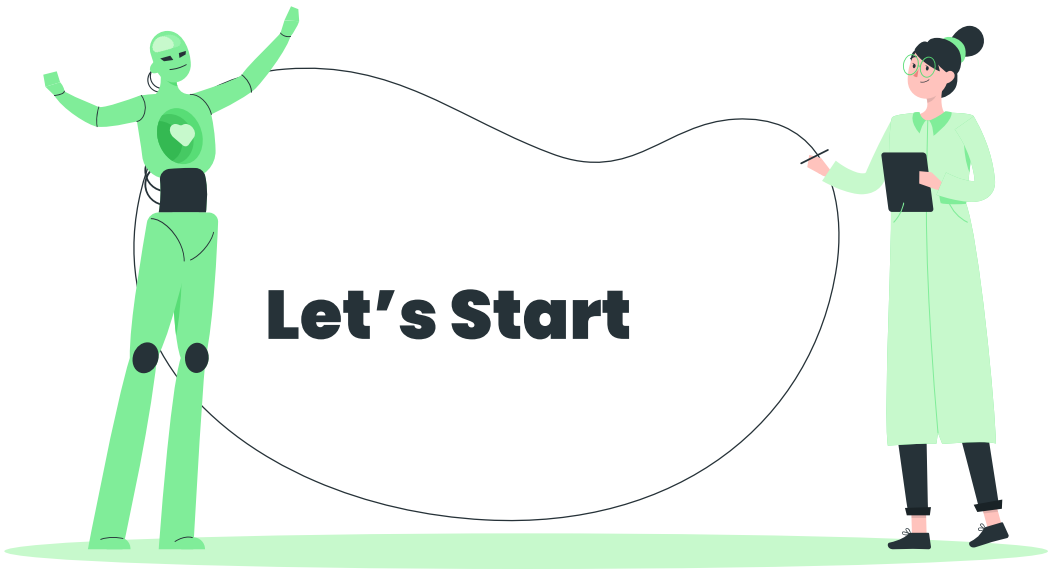
Robotics CS 3015

Umarah Qaseem

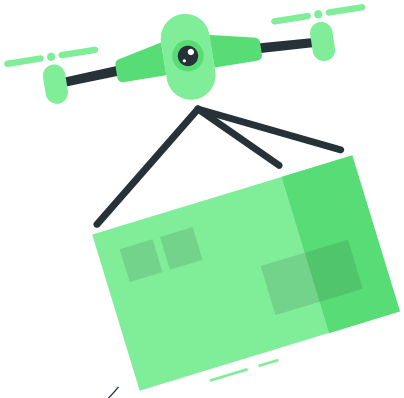


1



2

What is Robotics?



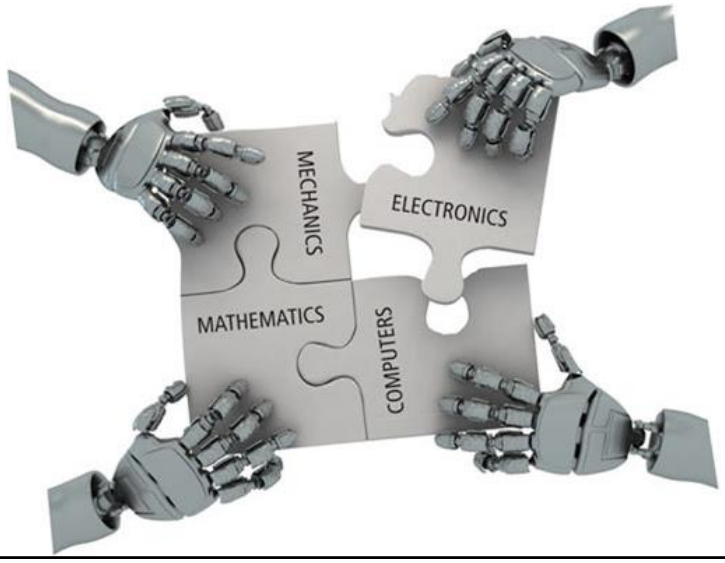
Robotics is an interdisciplinary field that involves design, construction, operation, and application of robots to perform tasks done traditionally by human beings.

It deals with the study of their control, sensory feedback and information processing.

Robotics – Umarah Qaseem

3

Elements of Robotics

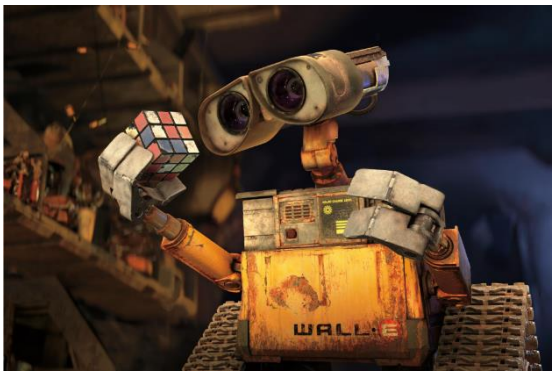


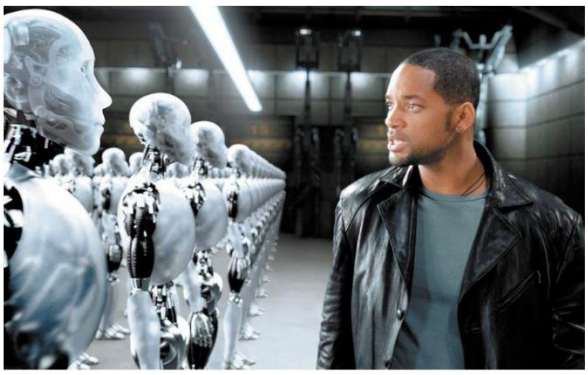
4

Robots in Fiction

5

Which movie?





Robotics – Umarah Qaseem

6

Star Wars 1977



7

Anthony Daniels – Actor



8



9

Laws of Robotics

1. **A robot may not injure a human being or, through inaction, allow a human being to come harm.**
2. **A robot must obey any orders given to it by human beings except where such where such orders would conflict with the First Law.**
3. **A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.**

Runaround, Isaac Asimov (1919-1992), Science Fiction and non-fiction author



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Robots in History

11

Automata 1808 – 1840





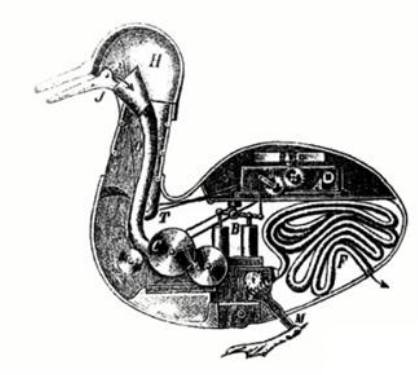
Photo: Daderot

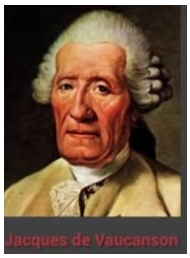


Maillardet's Automaton's Drawing of a ship

12

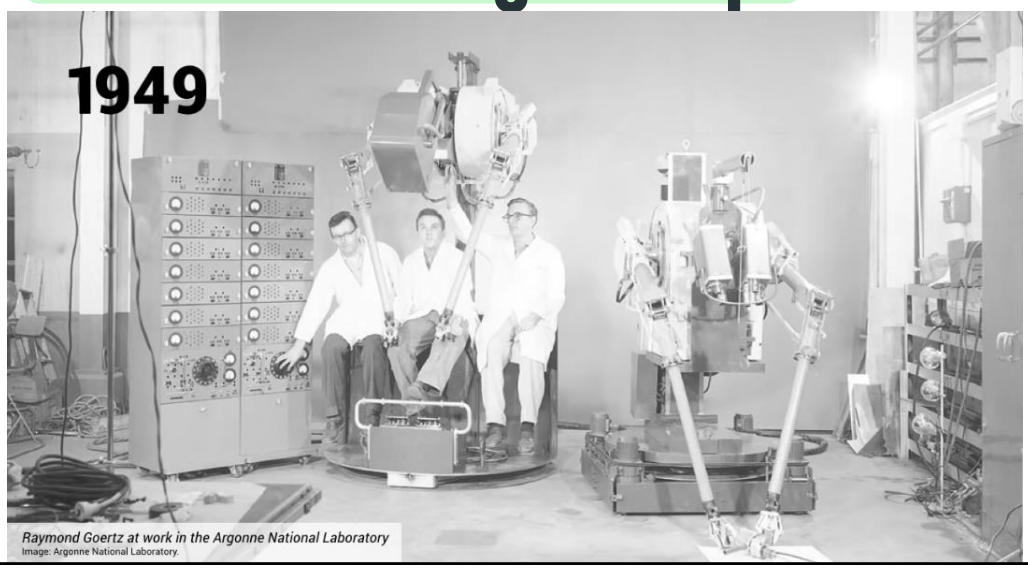
Digesting Duck (1739) – Vaucanson's Automata





13

Nuclear weapons – radioactive material handling – teleoperation



Raymond Goertz at work in the Argonne National Laboratory
Image: Argonne National Laboratory.

14

Unimation – 1956



George C. Devol
Image: The Estate of George C. Devol



Image: The Estate of George C. Devol



Joe Engelberger presenting at ARS Symposium 1996
Image courtesy of ARS Electronica

1956

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Robots Today

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More than 1M at work

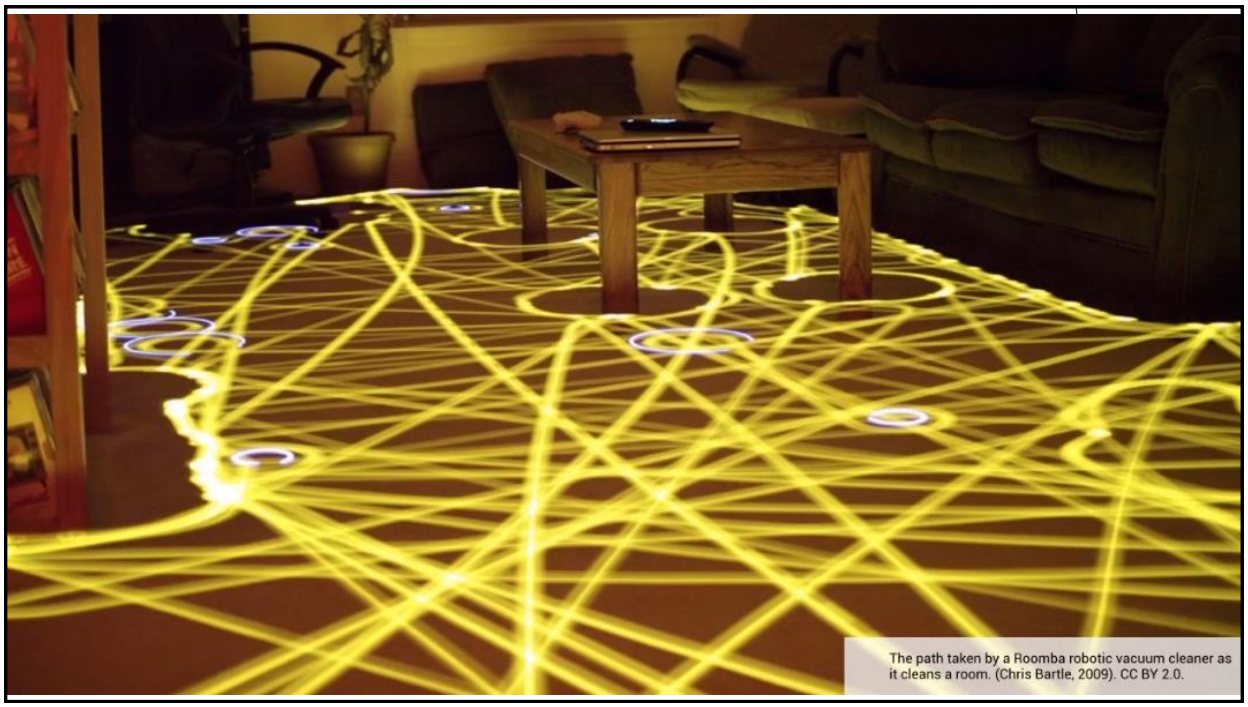


17

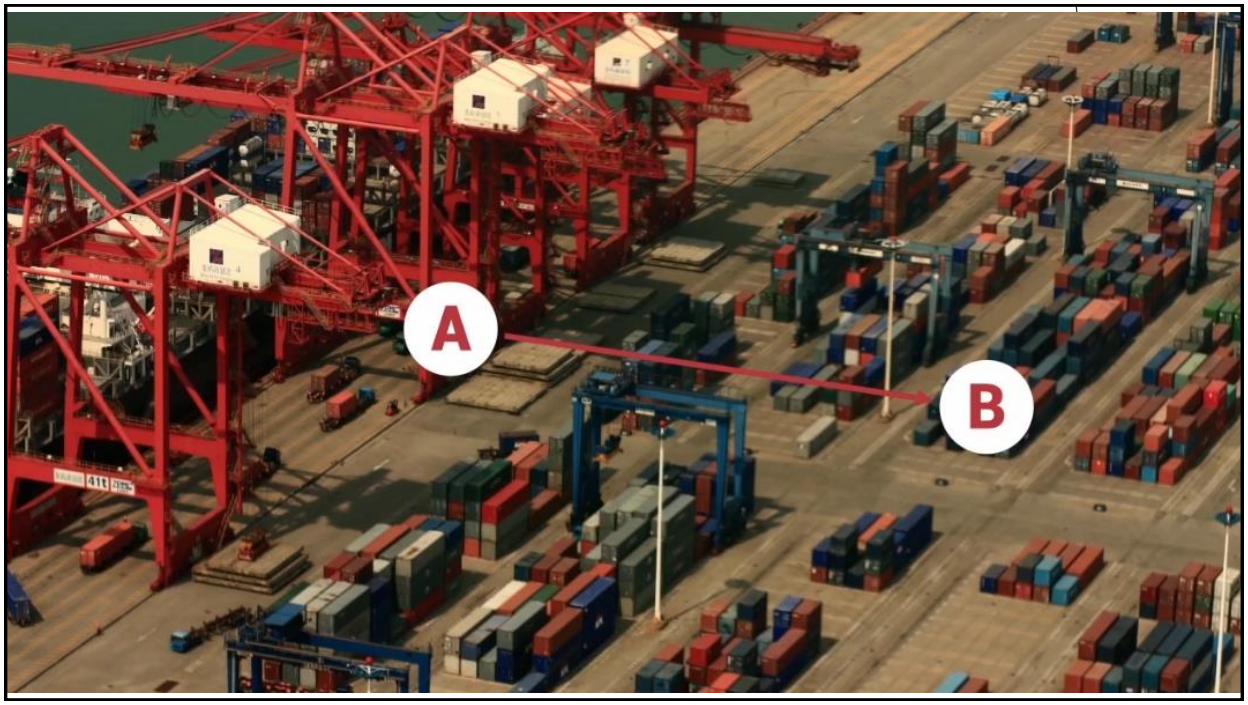
2004 – More than 10M sold



18


The path taken by a Roomba robotic vacuum cleaner as it cleans a room. (Chris Bartle, 2009). CC BY 2.0.

19


A
B

20

Second Generation of Robots

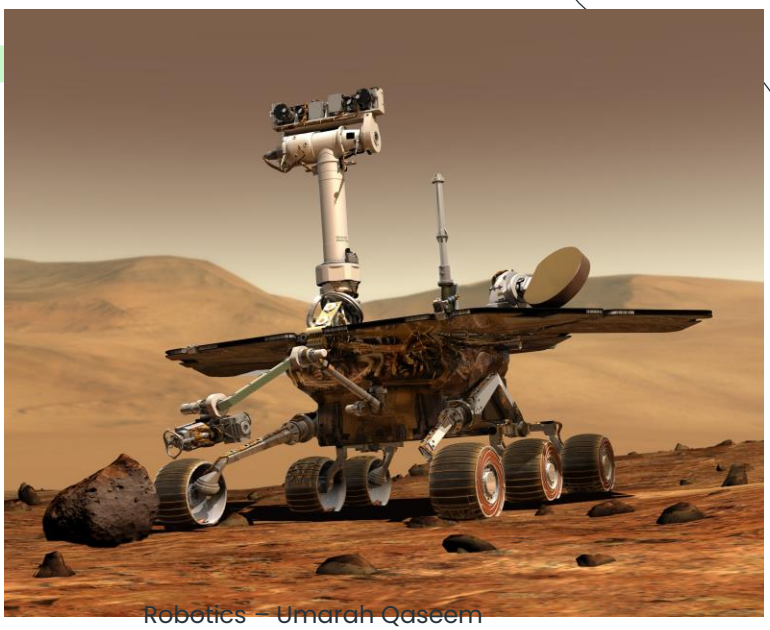
Little Dog:
https://www.youtube.com/watch?v=UlipbI0cAVE

Big Dog:
https://www.youtube.com/watch?v=cNZPRsrwumQ

Applications

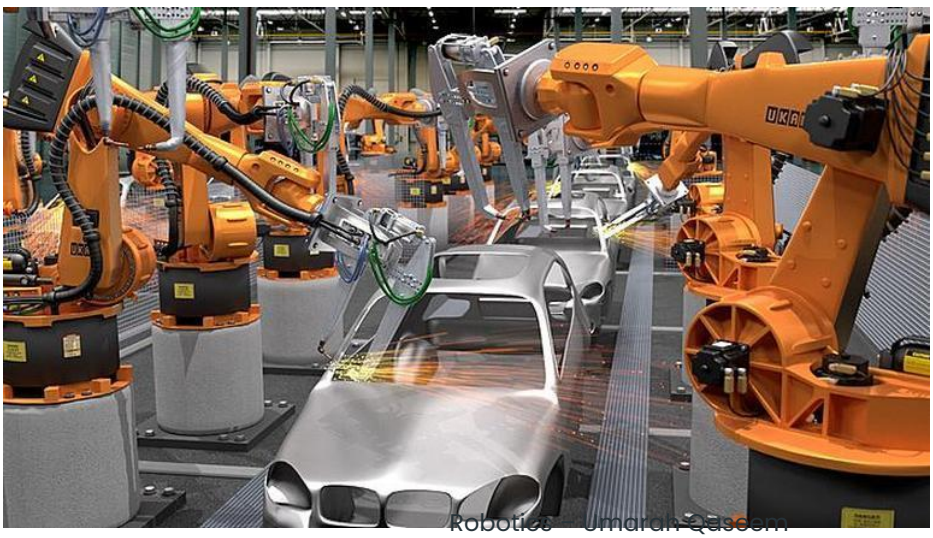
- Space Exploration and Missions
- Military
- Security and Surveillance
- Mining and Agriculture
- Healthcare
- Autonomous driving
- Domestic
- Industrial Automation

Mars Rover



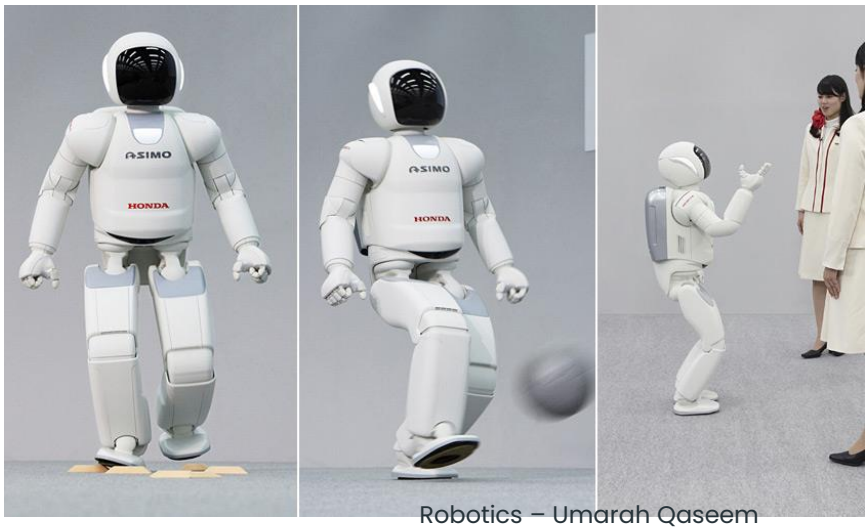
23

Industrial Robot



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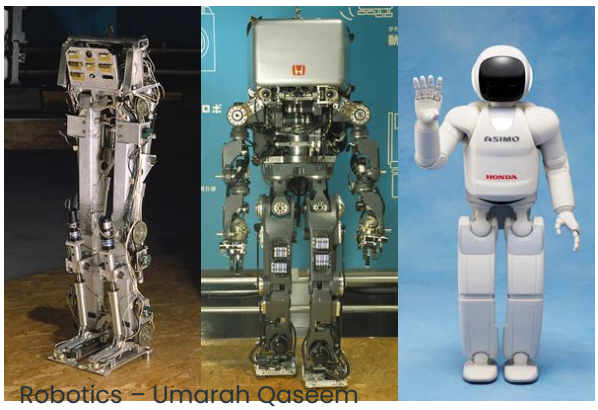
Humanoid – ASIMO



25

Humanoid – ASIMO





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Humanoid – Nao



27



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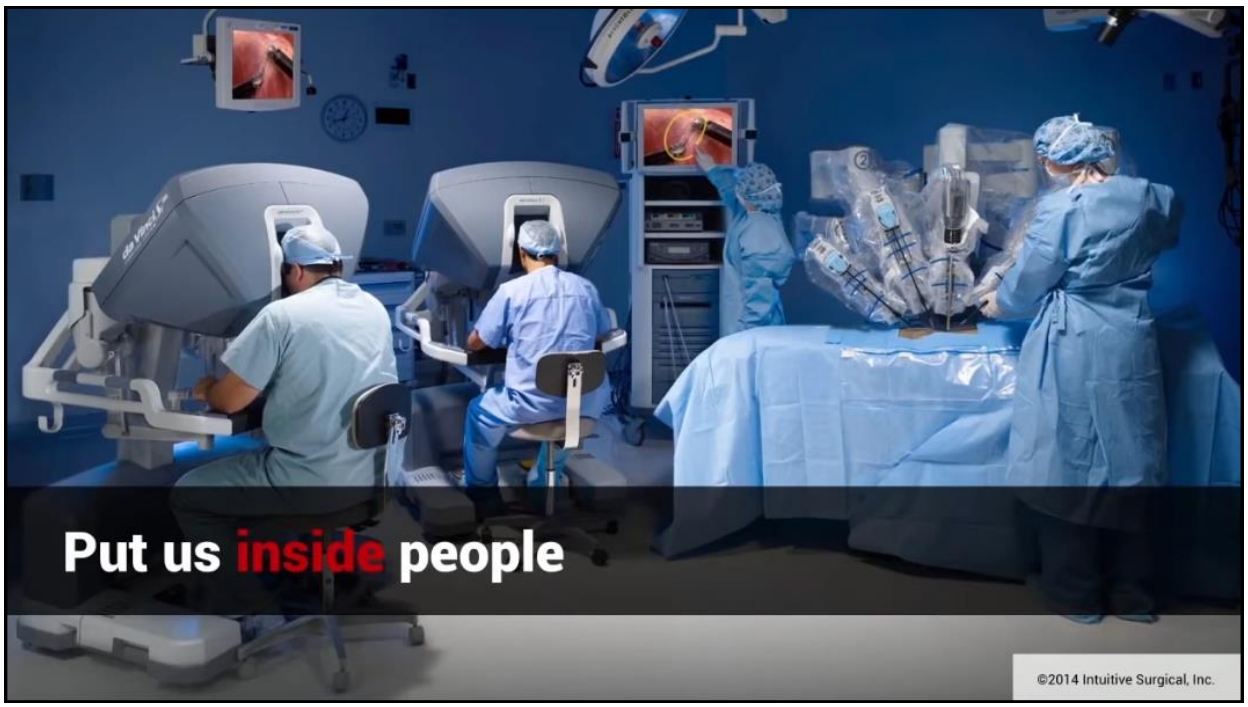


Shadow Robot Company | CC BY-SA 1.0

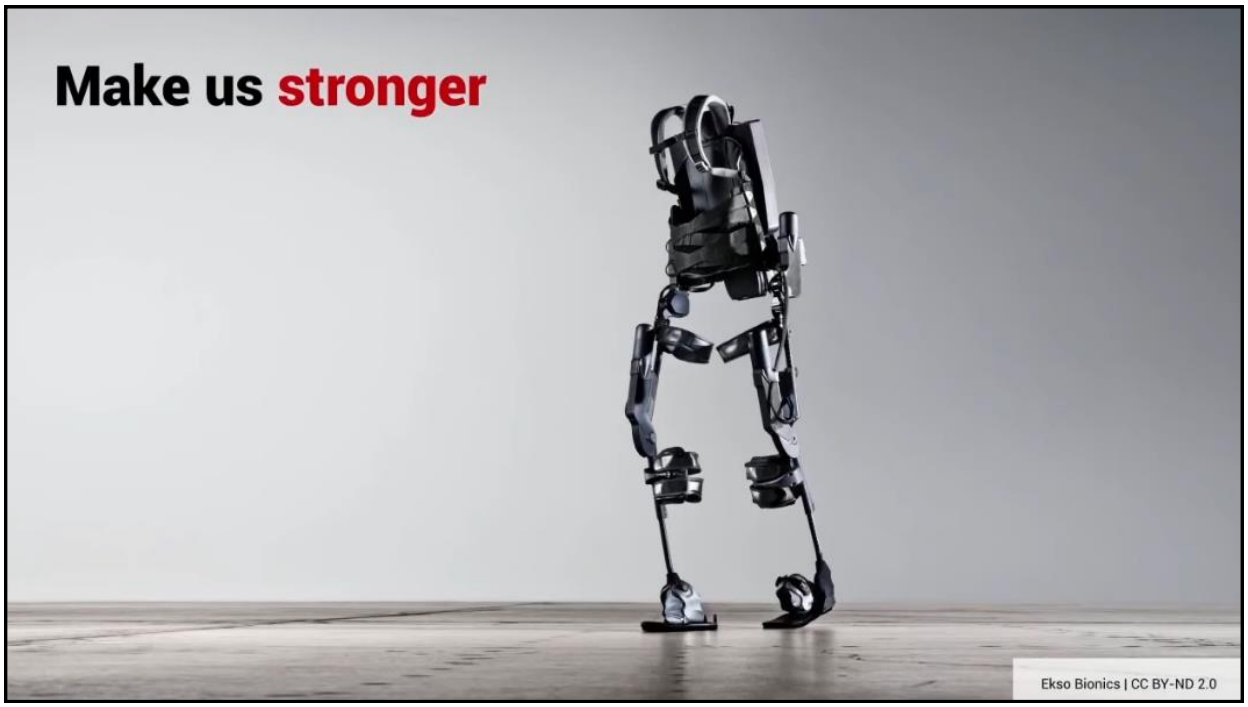


Shadow Robot Company | CC BY-SA 3.0

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30



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Challenge

- We know hands, vision systems, heads, legs, how to walk, how to handle objects etc.
- The big challenge today is how to integrate all this technology in a way that is
- reliable
- approachable
- and gives us a functionality approaching that of a human being.

Robotics – Umarah Qaseem

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Origin

- Term coined in a 1921 Czech Science fiction play "Rossum's Universal Robots" by Karel Capek.
- In Czech, the word "Robot" is derived from the Slavic word Rabota for slave.
- First patent for what we consider a robot was filed in 1954 by George C. Dovel and issued in 1961
- First Robotics company, Unimation was founded in 1956 by George C. Devol and Joseph Engelberger

The Word

- Modern term robot derives from the Czech word robota ("forced labour") used in Karel Čapek's play R.U.R. (1920).

- The play's robots were manufactured humans, heartlessly exploited by factory owners until they revolted and ultimately destroyed humanity.



Capek's Grave



Portrait of Capek

Father of Robotics

- Ismail al-Jazari is considered to be the father of robotics and modern engineering.



Father of Robotics

- George C. Devol

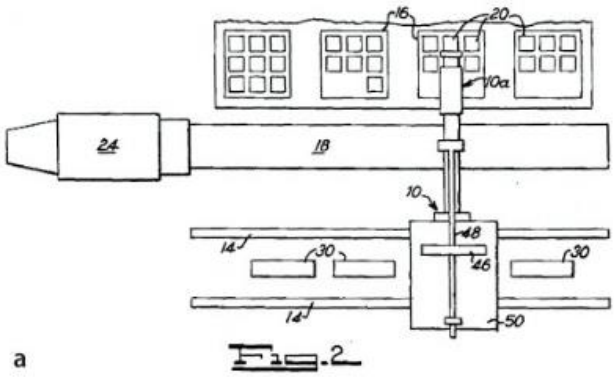


- Joseph F. Engelberger
- "Father of Robotics"



First Industrial Robot installed – 1961

- **a** A plan view of machine from Devol's patent; **b** first unimation robot working at a General Motors factory. (Courtesy of George C Dovel)





Descendants of Unimate

- A modern six-axis robot from ABB for factory automation (Image courtesy ABB robotics).



Descendants of Unimate

- Baxter, two-armed robot with built-in vision capability (image courtesy Rethink robotics)



Mobile Robots



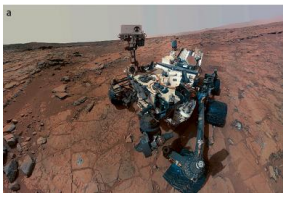
Small autonomous underwater vehicle (Todd Walsh © 2013 MBARI)



self driving car (image courtesy Dept. Information Engineering, Oxford Univ.)



Honda's Asimo humanoid robot (image courtesy Honda Motor Co. Japan)



Mars Science Lander, Curiosity, self portrait taken at "John Klein".



Cheetah legged robot (image courtesy Boston Dynamics)



Savioke Relay delivery robot (image courtesy Savioke)

Mobile Robots



Autonomous Underwater vehicle



Global Hawk unmanned aerial vehicle
(photo courtesy of NASA)

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What is the Future?

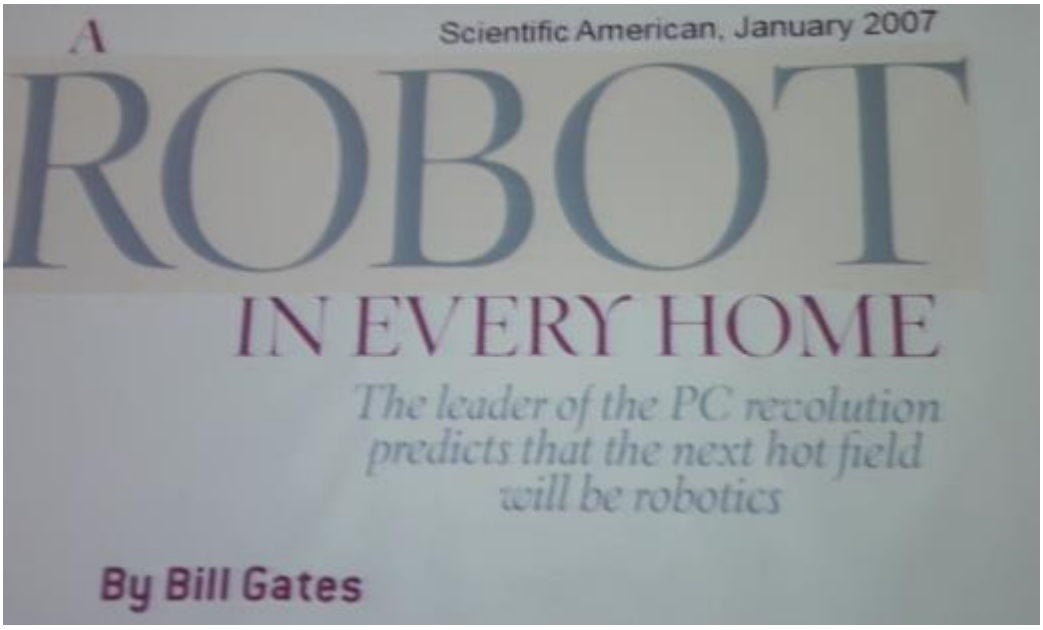
43

Main Technology Areas

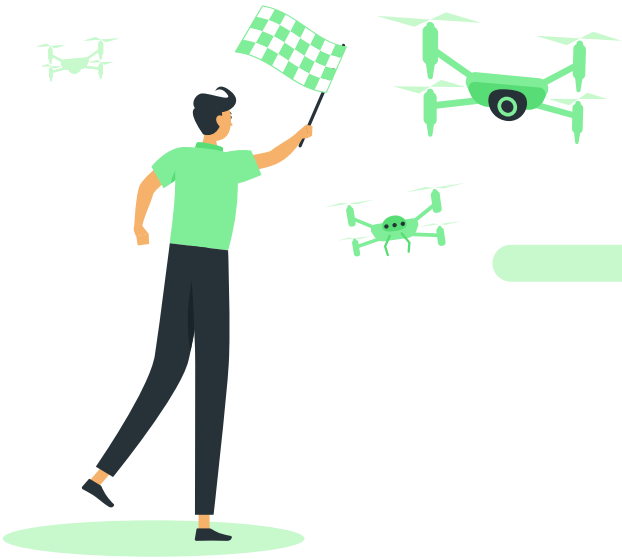
- Micro-electronics
- Computers
- Telecommunication
- ***Robotics***
- Biotechnology
- Green Energy



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45



Useful Field?

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Characteristics of Robots

- **Consistent**
- **Accurate**
- **Reliable**

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Characteristics of Robots

Do things that people

- **Can't do**
  - **Space, deep sea**
- **Won't do**
  - **Boring, dull**
- **Shouldn't do**
  - **Dangerous, unhealthy, risky**



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Is it a useful field?

It deals with automated machines that can take the place of humans in

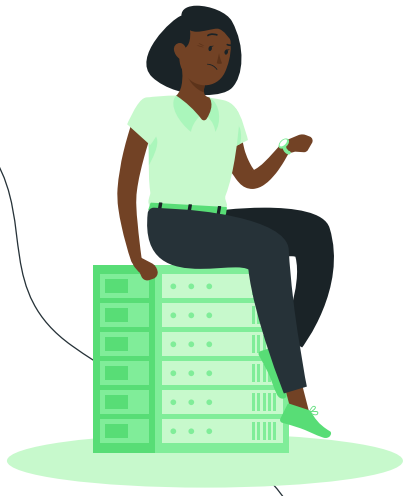
- Dangerous environments
  - inspection of radioactive materials
  - bomb detection and deactivation
  - finding survivors in unstable ruins, and exploring mines and shipwrecks
- Places where humans cannot survive
  - in space
  - Underwater
  - in high heat
  - clean up and containment of hazardous materials and radiation
- Manufacturing processes
- Laboratories, healthcare and complex medical surgery
- Commercial (welding, cutting, lifting, sorting and bending)
- Military

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Do you see a future of a Degree in Robotics?

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Let's Discuss

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Home Task

1) Install MATLAB before coming in next class

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